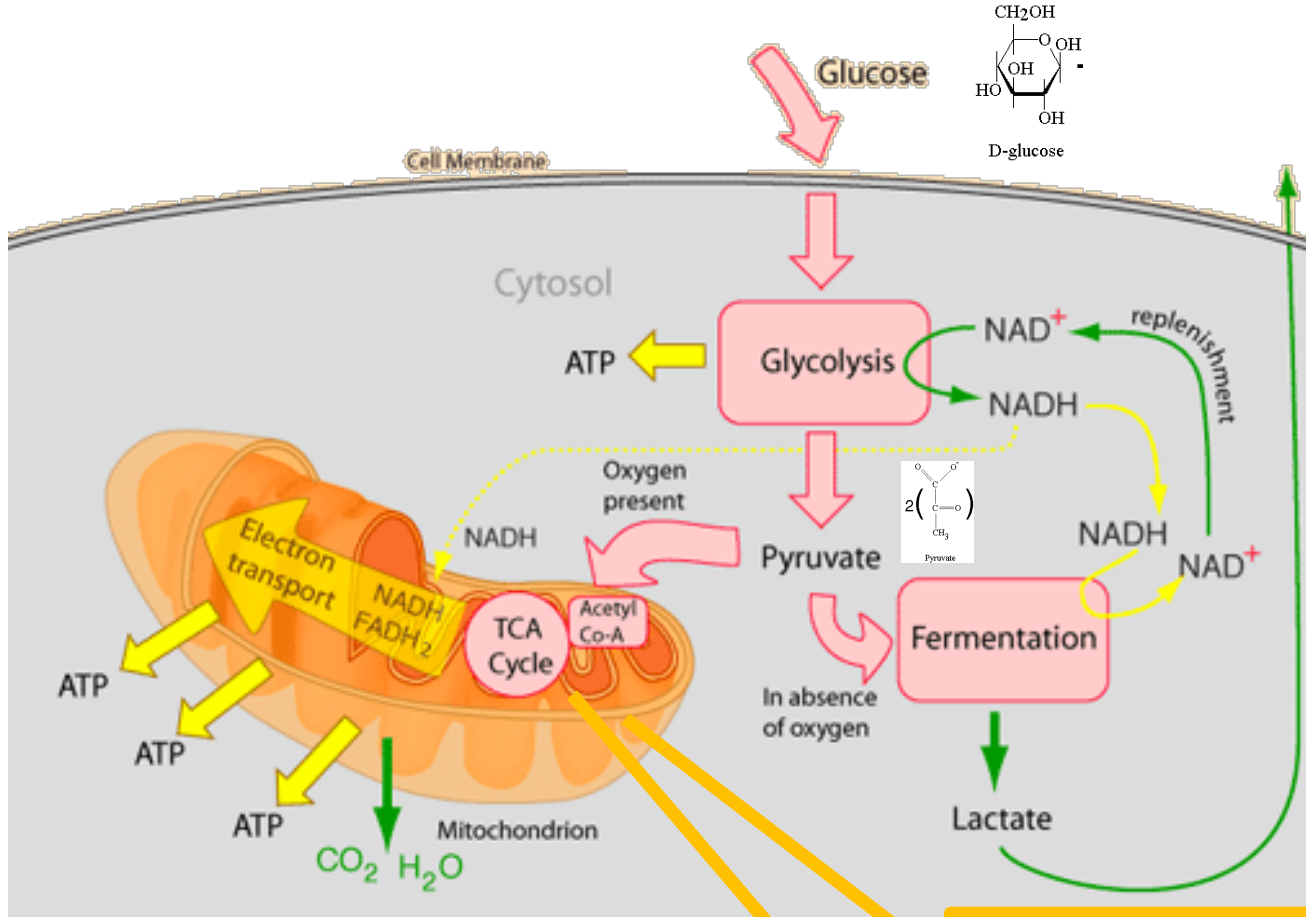
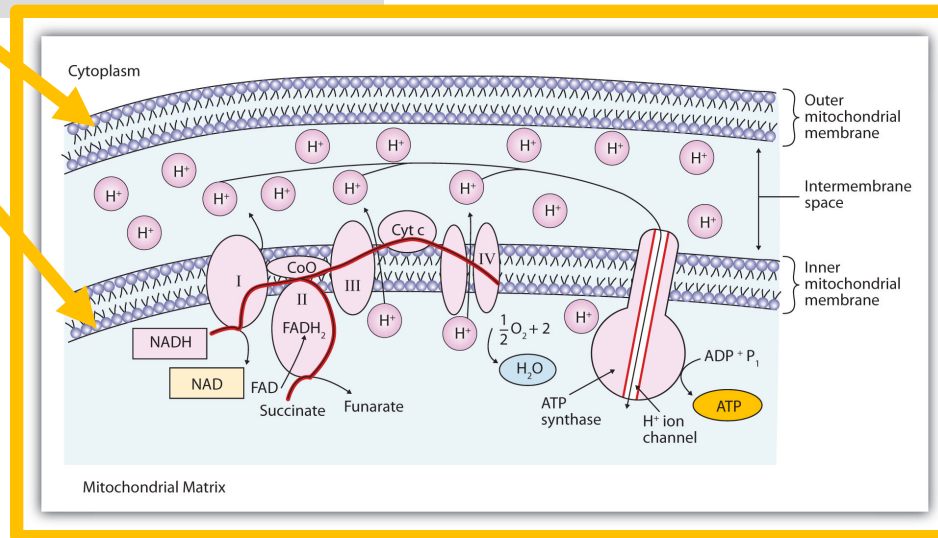
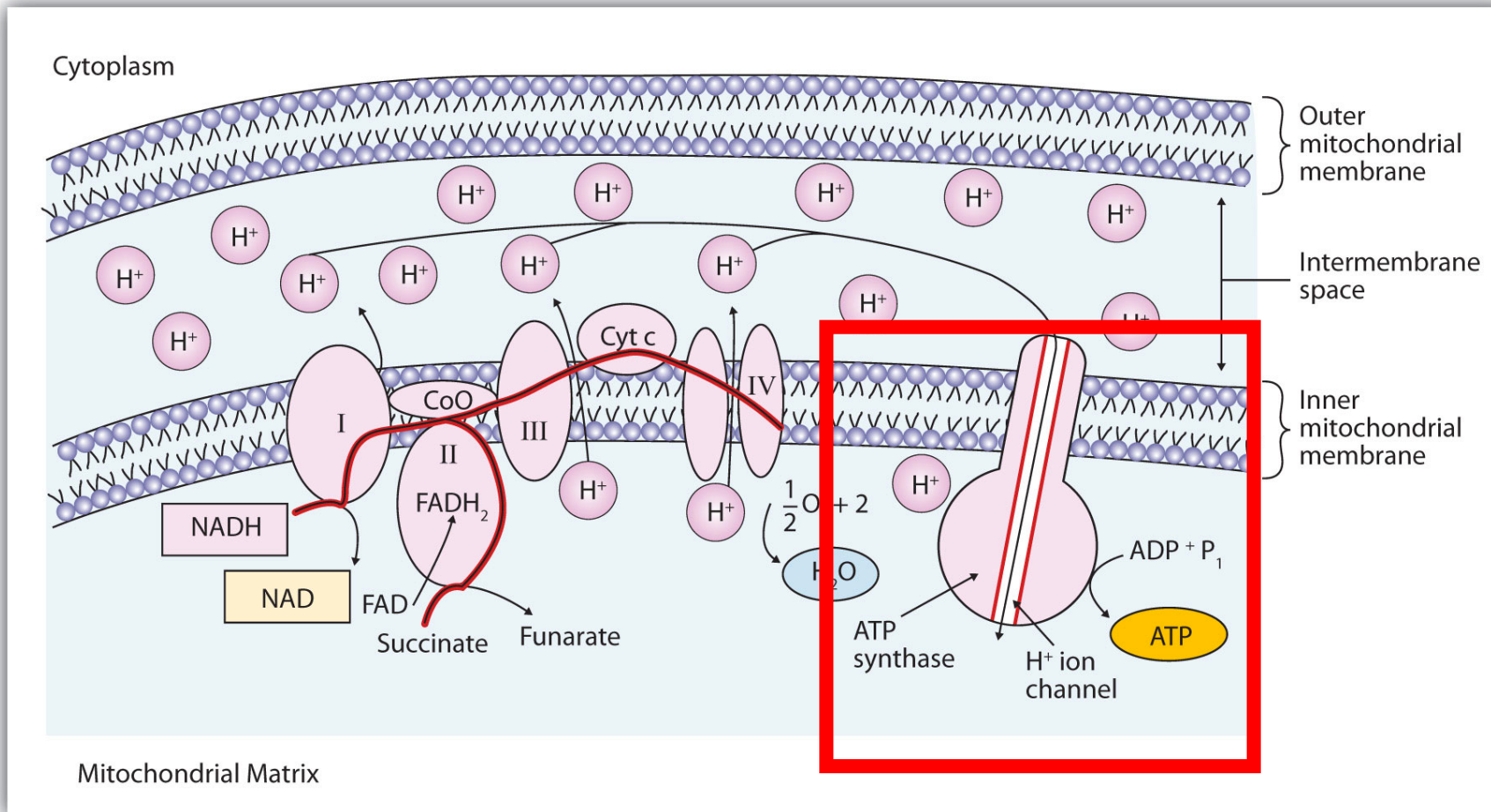




Recap.... Lecture 7



ATP synthase



Dr. P. Satpati,
BSBE, IIT Guwahati

INTER-MEMBRANE SPACE

ATP synthase = F0 + F1

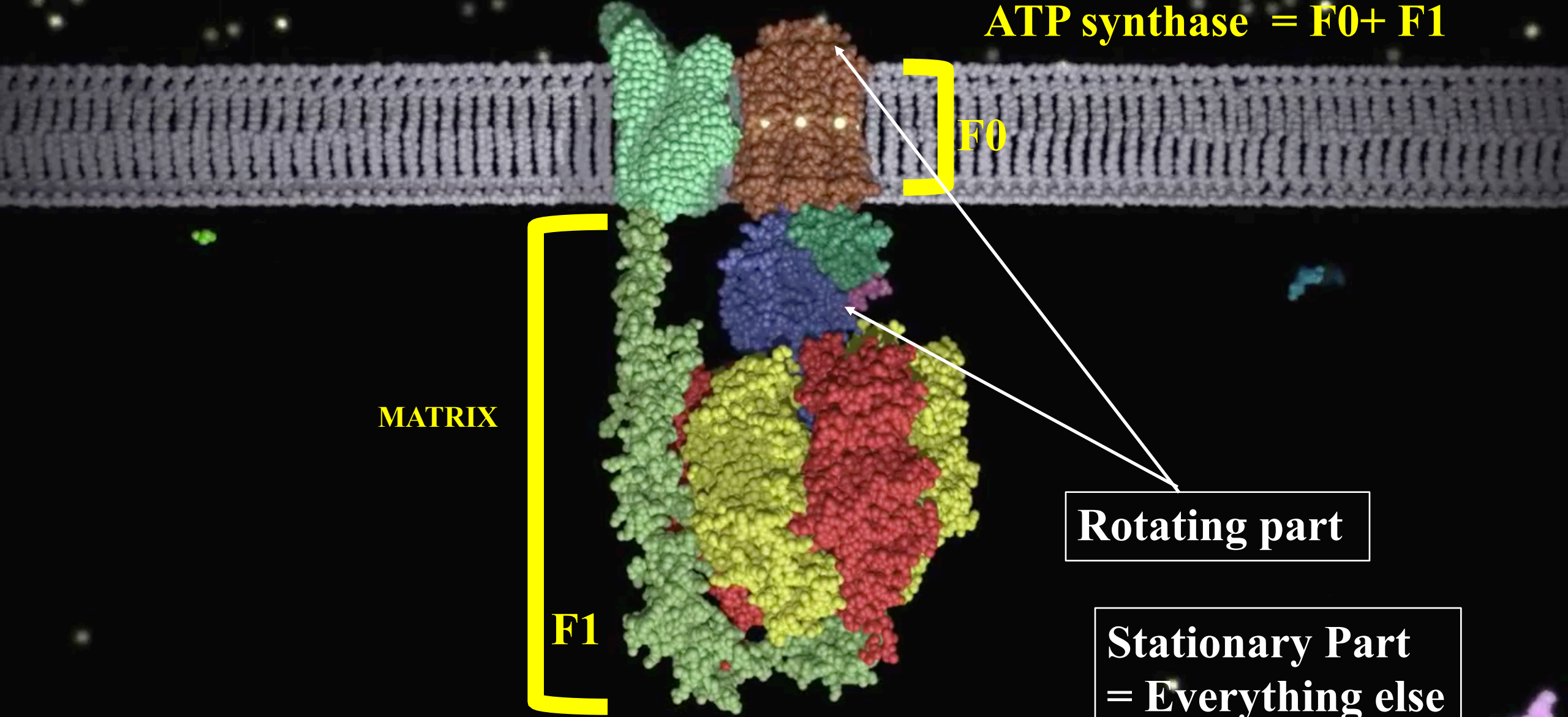
F0

MATRIX

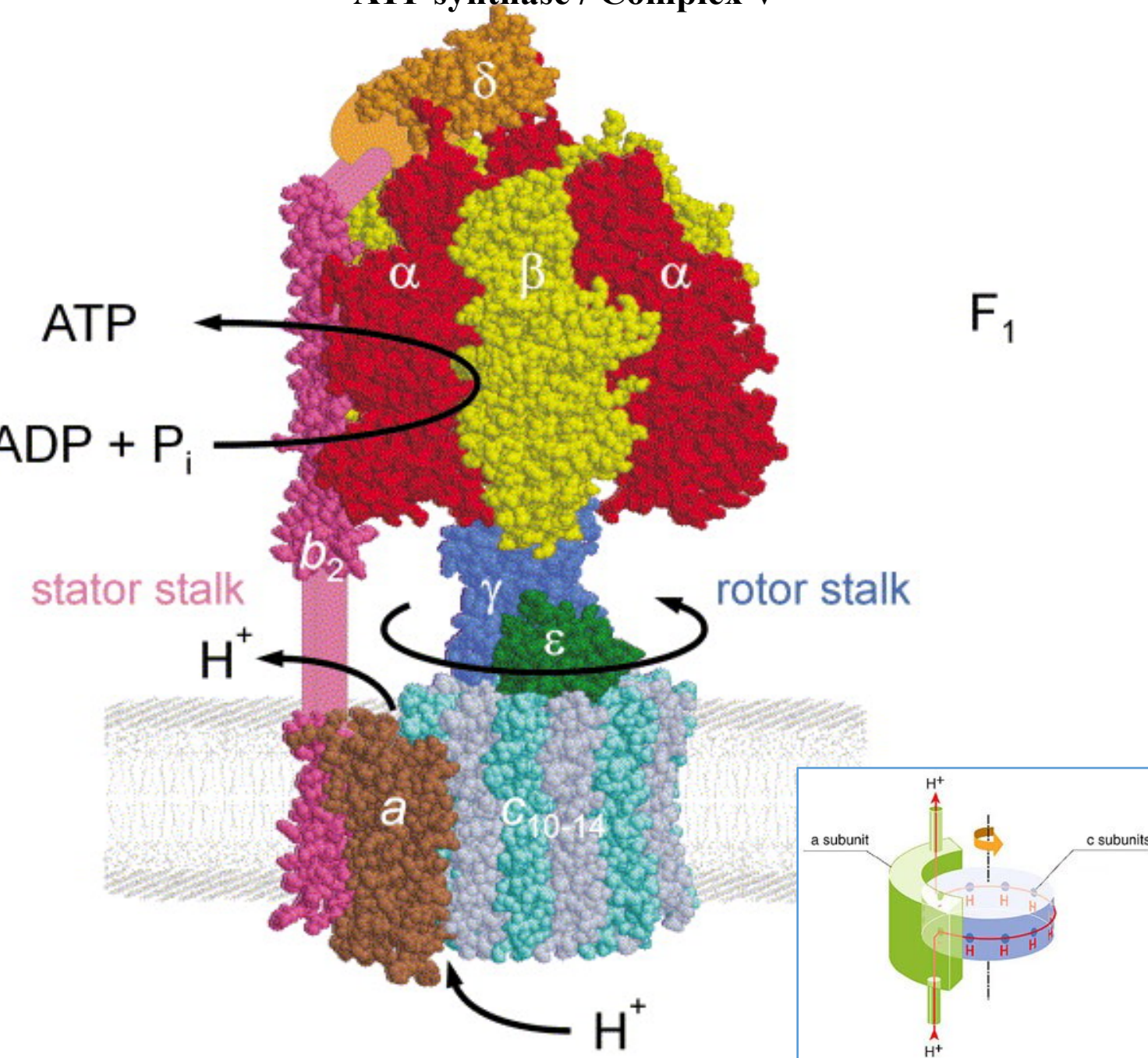
F1

Rotating part

**Stationary Part
= Everything else**



ATP synthase / Complex V

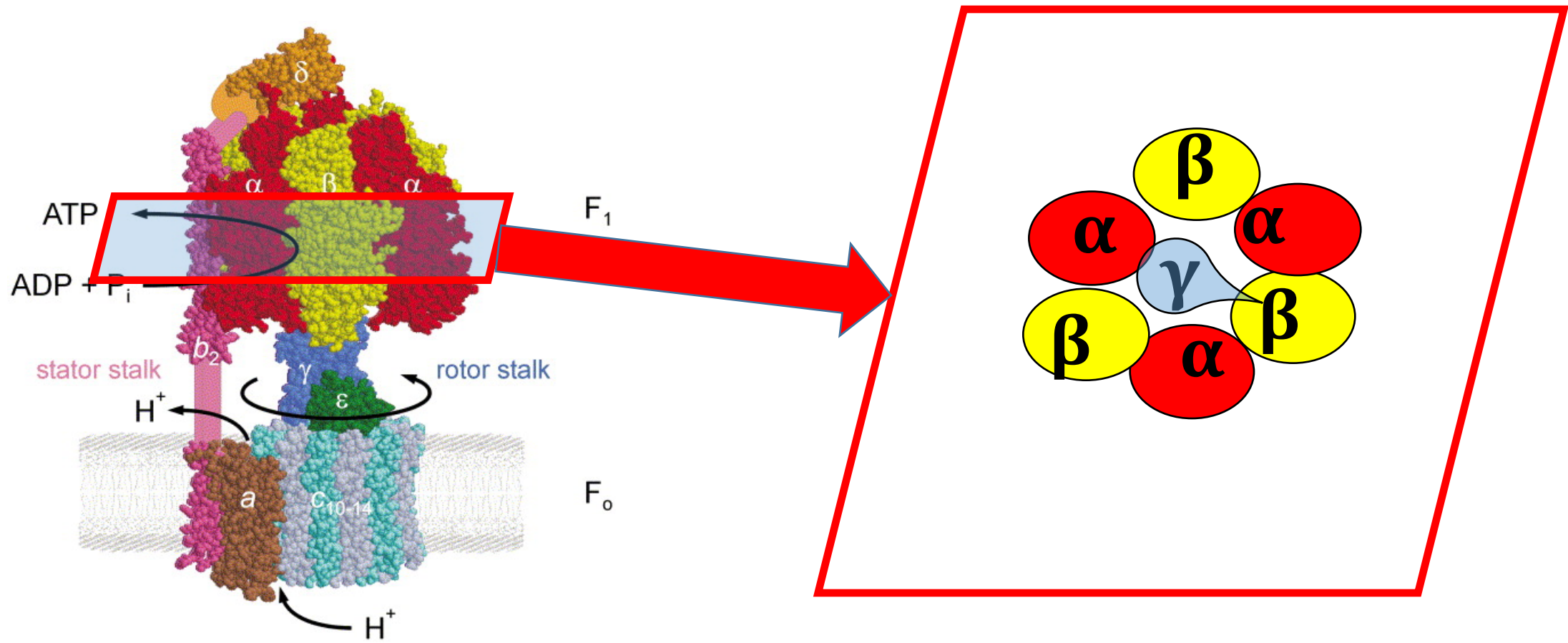


- Two regions F₀ and F₁ region
- F₁ region (Lies in the matrix of mitochondria)
 - 5 types (α , β , γ , ϵ , δ)
 - Hexameric $\alpha_3\beta_3$ ring (ATP synthesis takes place)
 - γ , ϵ organize to form central stalk/rotar stalk which runs through the inner cavity of the hexameric $\alpha_3\beta_3$ ring
 - δ holds $\alpha_3\beta_3$ hexameric ring in place. (stop rotation)
- F₀ region (Mostly hydrophobic and embedded in the inner membrane)
 - **10-14 "C" subunit organize into a ring. Acts as a proton channel and allow proton to flow.**
 - a single "a" subunit binds outside of the C ring. Helps connect the F₀ and F₁. Important for proton transfer

ATP synthase consist to:

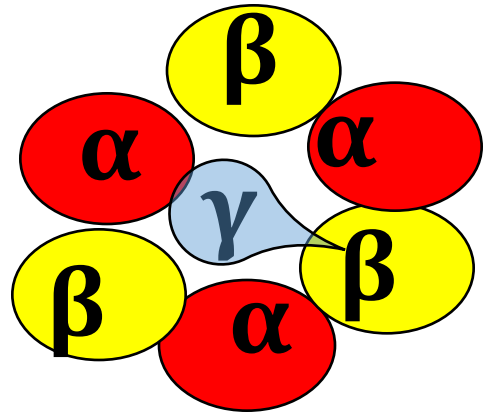
Rotatory region = C ring + $\gamma\epsilon$ rotar stalk

Stationary region = everything else



Remember:

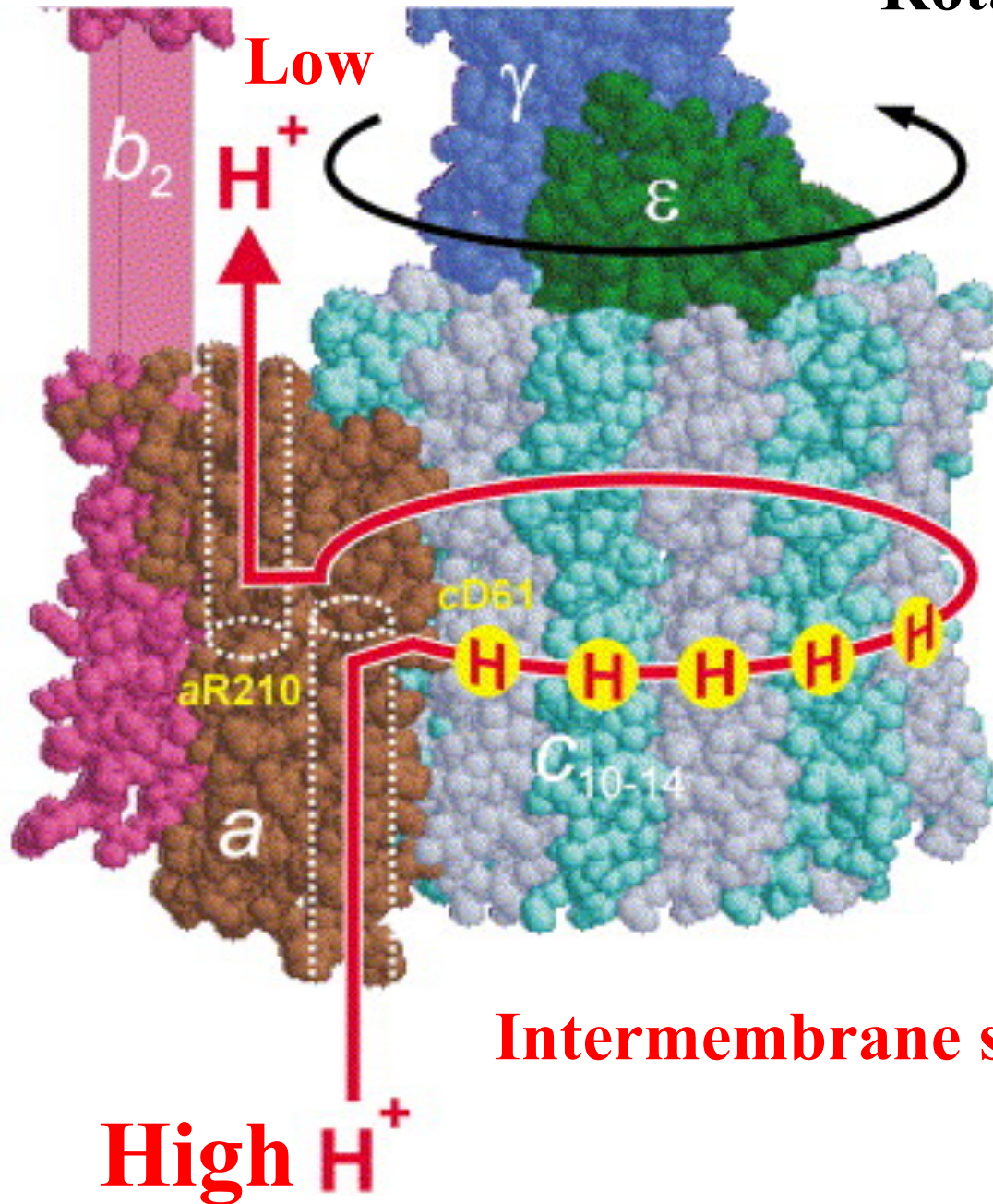
1. Rotar stalk γ
2. Rotation of γ stalk can induce conformational change in the α/β subunit



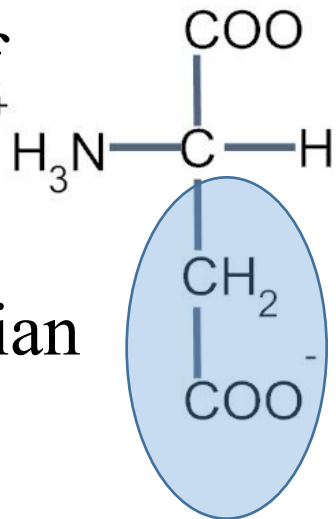
3 binding pockets (α/β interface)

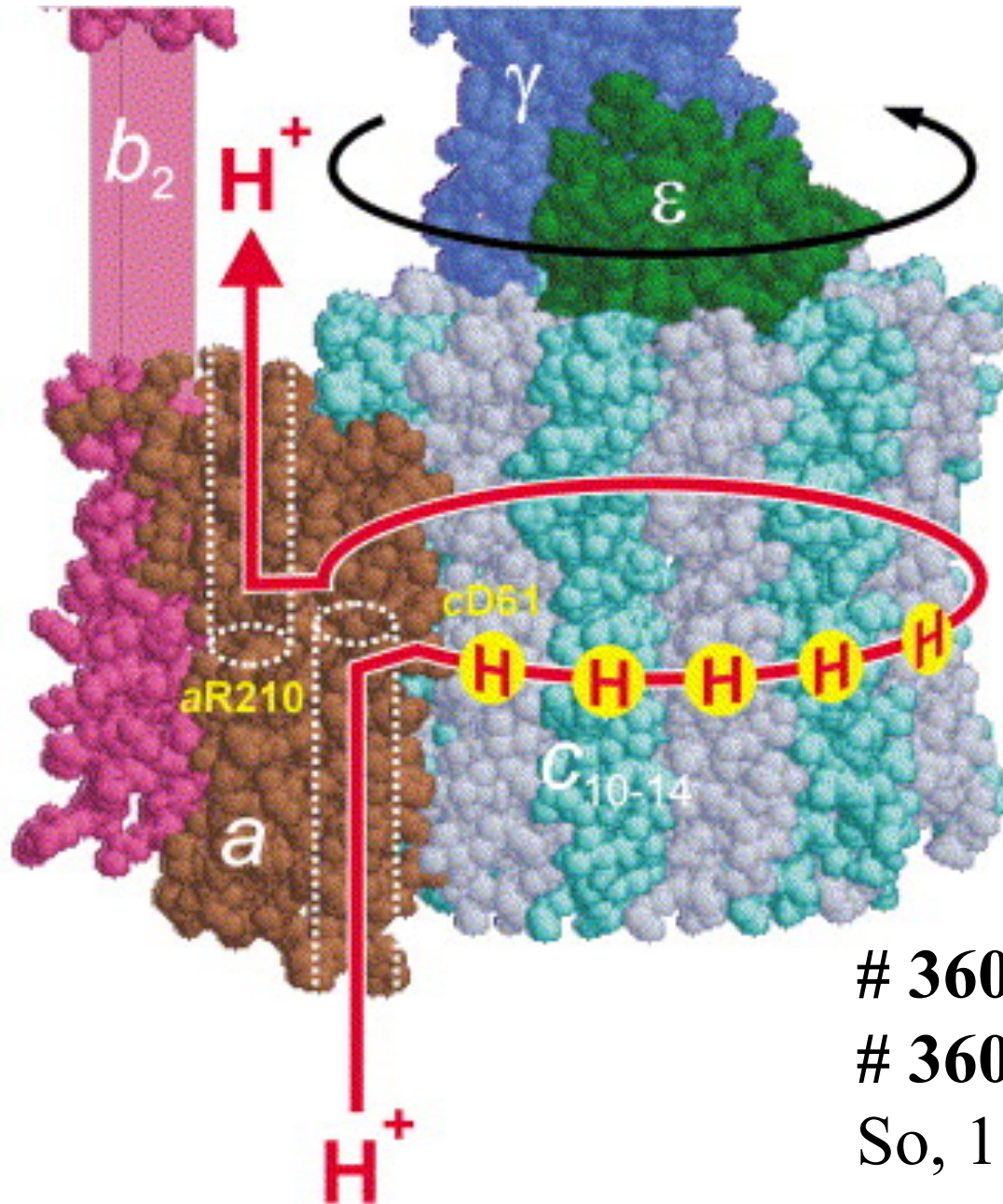
360° rotation of $\gamma = 3$ ATP molecule

Rotation ?



1. “a” subunit contain two hydrophilic halfway channel (DO NOT SPAN THE ENTIRE MEMBRANE).
2. Aspartate residue at the center of “c” subunit.
3. Mechanism (High $\rightarrow$ Low H⁺) through Asp protonation, Brownian motor.





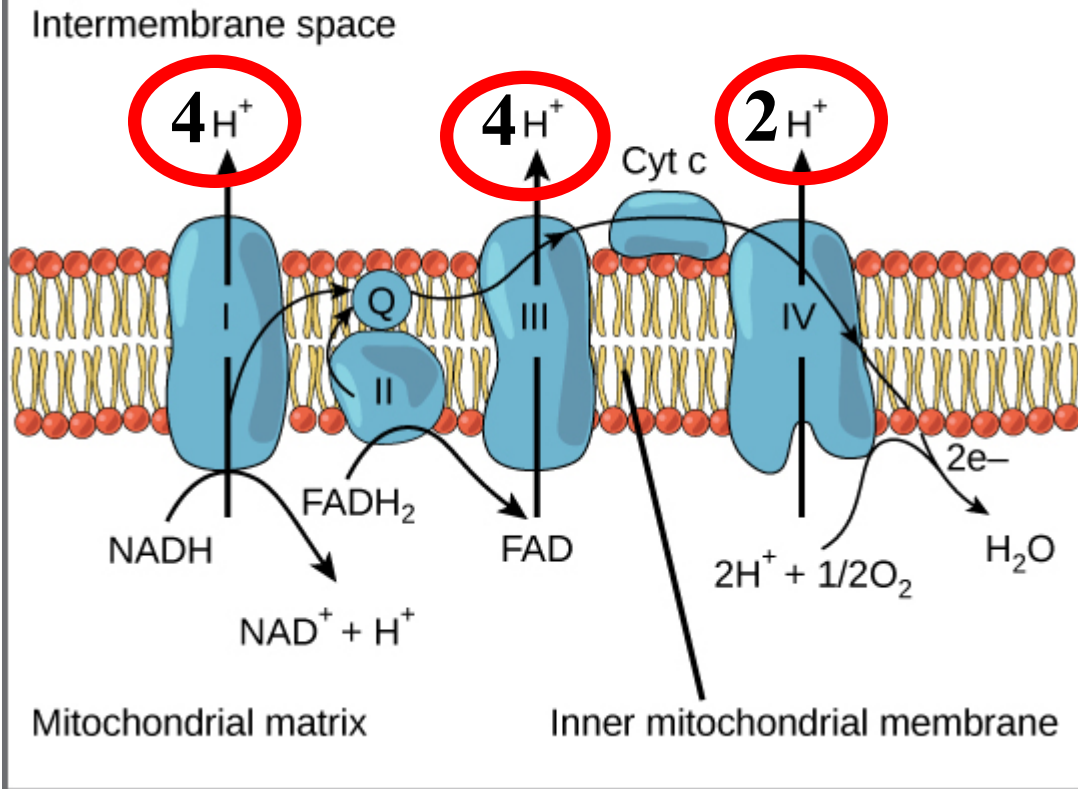
How many H⁺ diffusion required for 1 ATP ?

360° rotation = 3 ATP

360° rotation = 10-14 H⁺ (**remember C10-14**)

So, 1 ATP = 3.33 – 4.67 H⁺ ~ 4H⁺

Electron Transport Chain



1 ATP production = 4 H⁺

1 NADH = 10 H⁺ = 2.5 ~ 3 ATP

1 FADH₂ = 6 H⁺ = 1.5 ~ 2 ATP

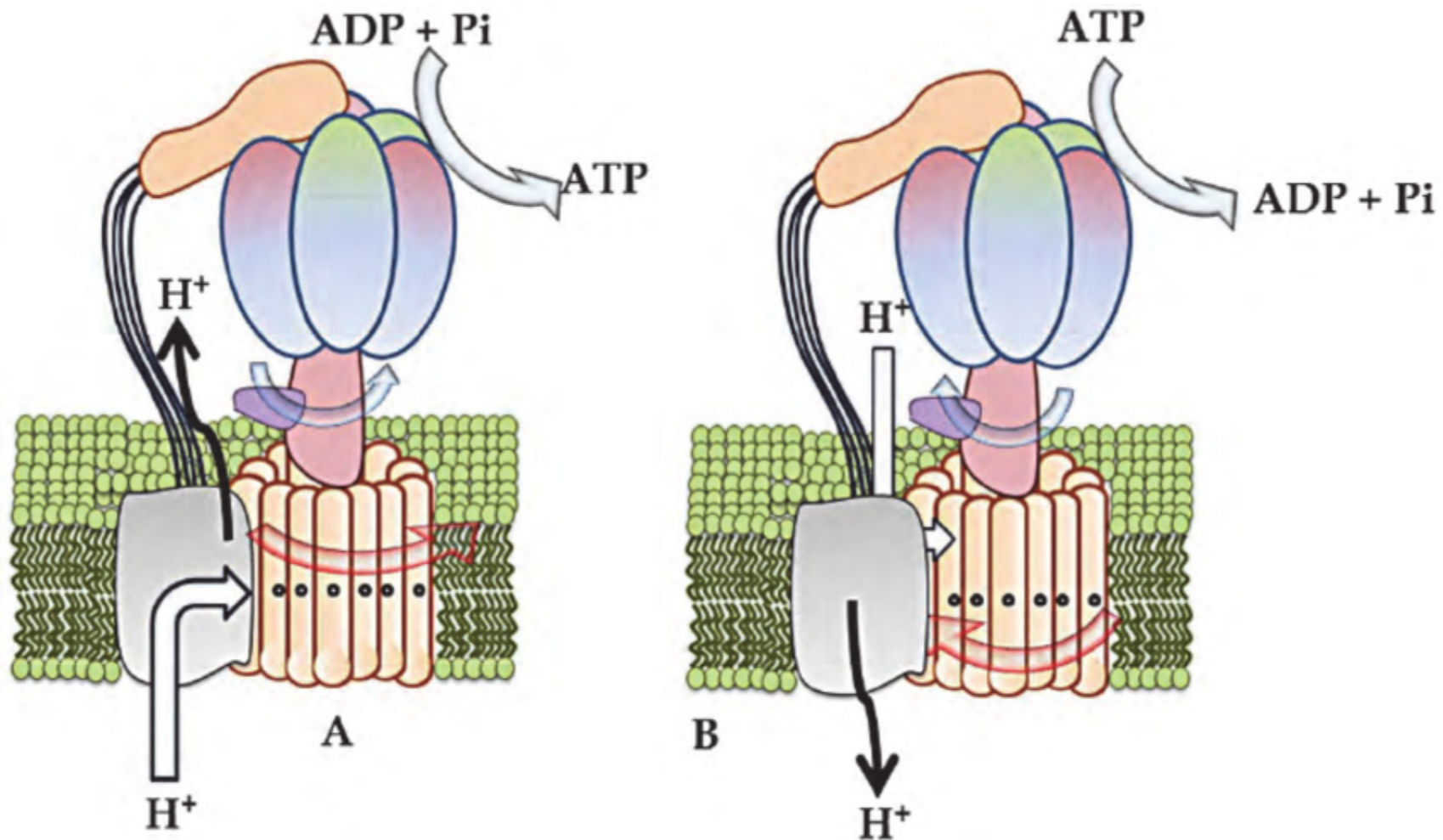
We have used this
for calculating Total ATP
production



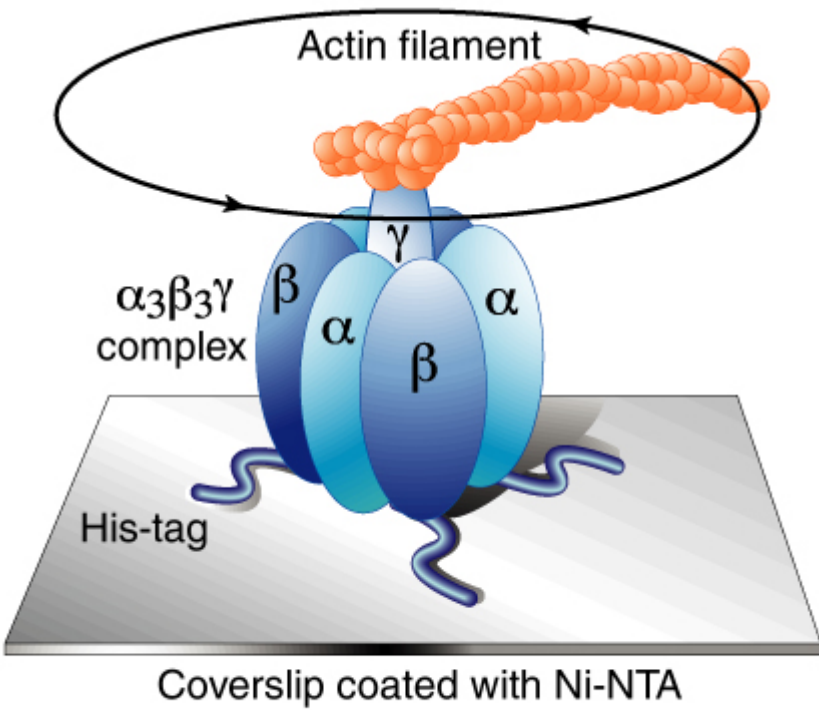
ATP synthase

Stock Lab
The Victor Chang Cardiac Research Institute

Rotation can happen in both the direction



https://www.researchgate.net/figure/Schematic-representation-of-ATP-synthesis-and-hydrolysis-by-F-0-F-1-ATP-synthase-A_fig2_221924465



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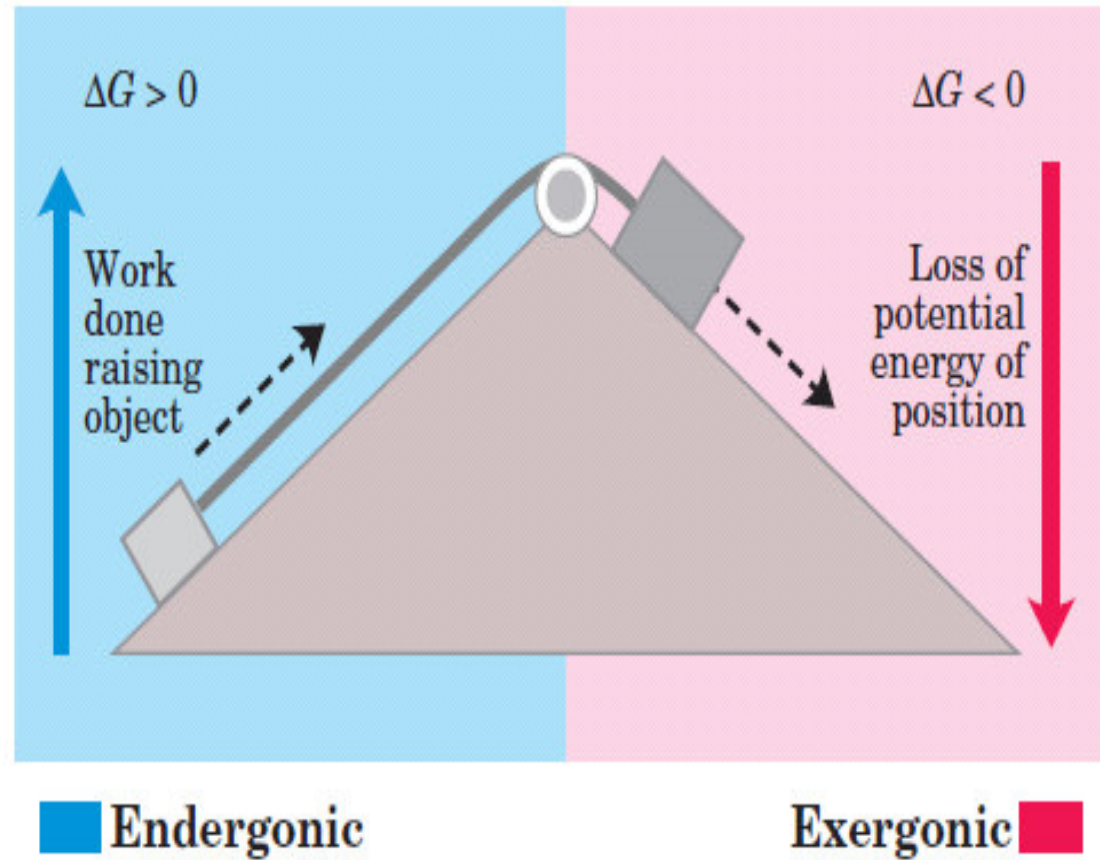
<https://www.youtube.com/watch?v=0y7n-vK1AJE>

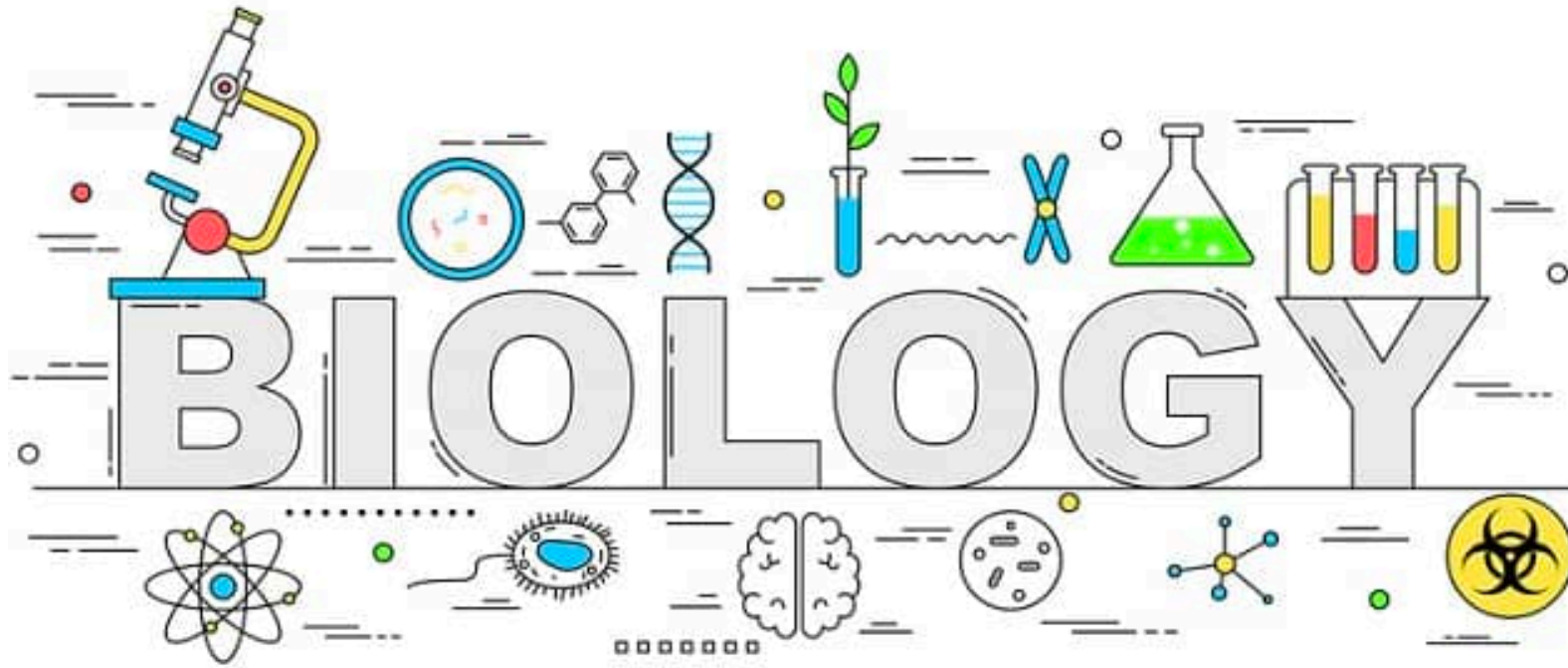
F1 ATPase

Credit: Kazuhiko Kinosita, Waseda University, Japan

Life is all about energy coupling

Energy Coupling:





Next: Thermodynamics and ATP synthesis